

ANALYSIS REPORT

Total Swap Attempts : 3,877,013 (100.0%)

Completed Swaps : 3,645,809 (94.04%)

Unsuccessful Swaps : 231,204 (5.96%)

Unsuccessful Swap Attempt Breakdown

Failure Classification	Event Count	Share of Total Failures (%)	Core Operational Driver
failed_no_charged_battery	134,389	58.12%	High demand outpacing station battery recharge rate
abandoned_queue	68,533	29.64%	Excessive wait times during peak station congestion
cancelled_by_rider	16,712	7.23%	Pre-swap rider cancellation
failed_system_error	11,570	5.00%	Station firmware, BMS, or socket communication failure

2. Temporal Analysis & Failure Volatility

Monthly Degradation & Summer Spikes

Baseline failure rates remain stable between **4.20% and 4.40%** during standard operational months. Significant operational failure spikes occur during extreme heat months (May–June):

- **May 2024:** Failure rate peaked at **12.27%** (17,950 failed attempts).
- **June 2024:** Failure rate remained high at **10.17%** (15,145 failed attempts).
- **May 2025:** Failure rate spiked again to **9.90%** (33,851 failed attempts).
- **June 2025:** Failure rate recorded at **8.55%** (28,738 failed attempts).

Technical Insight: High ambient summer temperatures slow safe lithium-ion charging cycles due to BMS thermal throttling, causing stations to run out of fully charged batteries during peak demand hours.

Hourly Peak Load & Queue Abandonment

- **Evening Surge Window (19:00 – 21:00):** Station utilization peaks with over 447,000 attempts per hour, pushing queue abandonment rates to their highest daily levels (**1.93% – 1.99%**).
- **Midday Surge Window (12:00 – 13:00):** Secondary demand peak recording $\approx$ 376,000 swap attempts.
- **Queue Wait Time Impact:** Riders who abandoned queues waited an average of **702.5 seconds ($\approx$ 11.7 minutes)** before leaving.

3. Geographic & Station Level Breakdown

Regional Performance Matrix

City	Total Attempts	Completed Swaps	Failure Rate (%)	Uncharged Battery Failures
Jaipur	487,634	449,366	7.85%	23,303 (4.78% of city total)
Delhi NCR	768,913	712,385	7.35%	33,892 (4.41% of city total)

City	Total Attempts	Completed Swaps	Failure Rate (%)	Uncharged Battery Failures
Hyderabad	711,930	664,312	6.69%	28,274 (3.97% of city total)
Pune	552,910	525,331	4.99%	15,480 (2.80% of city total)
Bengaluru	825,327	787,697	4.56%	20,602 (2.50% of city total)
Mumbai	530,299	506,718	4.45%	12,838 (2.42% of city total)

Worst Performing Stations

The top 10 highest-failure stations are concentrated exclusively in **Delhi NCR** and **Jaipur**, with failure rates ranging between **8.61%** and **9.08%**:

1. **STN-DEL-037** (Delhi NCR): 30,838 attempts → **9.08% failure rate**
2. **STN-DEL-048** (Delhi NCR): 34,646 attempts → **8.98% failure rate**
3. **STN-JAI-137** (Jaipur): 44,371 attempts → **8.75% failure rate**
4. **STN-JAI-142** (Jaipur): 33,319 attempts → **8.75% failure rate**
5. **STN-JAI-141** (Jaipur): 45,594 attempts → **8.73% failure rate**

4. Battery Fleet Degradation & Vendor Quality Audit

Fleet Health Overview

- **Total Active Fleet:** 6,500 battery packs.
- **Average Initial SOH:** 99.19%.
- **Average Current SOH:** 81.01%.

Vendor Audit Summary

Supplier Name	Total Pack Count	Avg Initial SOH	Avg Current SOH	Performance Classification
Amptek	1,307	99.19%	81.01%	Normal linear degradation
Cellora	3,555	99.18%	80.93%	Normal linear degradation
Kyron	1,638	99.21%	63.75%	Defective Manufacturing Lots

Isolation of Defective Batches (Kyron)

A lot-level degradation analysis revealed a manufacturing defect confined to three specific Kyron production lots:

- **KY-2407:** 480 total packs → **476 retired End-of-Life** (Avg SOH: 60.99%).
- **KY-2408:** 500 total packs → **489 retired End-of-Life** (Avg SOH: 61.05%).
- **KY-2409:** 481 total packs → **473 retired End-of-Life** (Avg SOH: 60.92%).

Defect Summary: A total of **1,438 Kyron battery packs** failed prematurely, reducing operational fleet capacity. Other Kyron manufacturing lots (KY-2410 through KY-2418) function normally with an average SOH of **~86.5%**.

5. Telemetry & Data Quality Issues Logged

1. **Negative Odometer Values:** Identified 7,814 event records containing negative values for km_since_last_swap (e.g., -19.1 km, -18.9 km) due to CAN bus telemetry sync corruption.
2. **BMS Calibration Drift:** Identified 3,356 swap events logging SOH > 100% (up to 101.4%), indicating uncalibrated BMS current integration sensors.
3. **Missing Exit Metrics on Failures:** Unsuccessful swap records (231,204 rows) systematically omit battery_out_id, soc_out_pct, soh_out_pct, and energy_to_recharge_kwh as no outgoing battery was dispensed.

Strategic Action Plan

1. **Vendor Reclamation & Legal Claim:** Issue a formal warranty claim to **Kyron** for the **1,438 defective packs** across lots KY-2407, KY-2408, and KY-2409.

2. **Thermal Mitigation Strategy:** Retrofit station active-cooling enclosures in **Jaipur** and **Delhi NCR** to maintain fast charging throughput during May–June ambient heat spikes.
3. **Peak Dynamic Rebalancing:** Implement automated battery transfer trucks to rebalance inventory at **Commercial Hub** and **Market** stations during the 18:00–21:00 peak demand window.